Effect-direction of noncoding regulatory variants is not predictable from sequence because chromatin accessibility and gene expression decouple: a benchmark and mechanistic analysis in human microglia

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Abstract

Background. Sequence-to-function deep-learning models — DeepSEA and Basset in the first generation, and the long-context models Borzoi and AlphaGenome today — predict genomic assay tracks with high accuracy and are increasingly used to interpret the noncoding variants that underlie most disease-associated genetic risk. Their utility for variant interpretation rests on a harder question than track prediction: can they predict the *direction* (sign) of a variant's regulatory effect in a defined cell context? This quantity is rarely benchmarked directly.

Results. We assembled a chromosome-split benchmark of noncoding variants with laboratory-measured effect direction spanning nine cellular contexts (6,377 variant×context measurements), and added a native human-microglia chromatin-accessibility QTL (caQTL) map from 95 donors as an on-target test for both frontier models. On the fairest achievable test — chromatin-accessibility direction in the native cell type — AlphaGenome (accuracy 0.537, 95% CI 0.485–0.587), Borzoi (0.551, 0.499–0.601), motif-PWM floor baselines, and models trained in-distribution on the caQTL data itself were all statistically indistinguishable from chance (majority-class baseline 0.565). To explain this, we tested whether a variant's effect on chromatin accessibility predicts its effect on gene expression in the *same 95 donors* (caQTL versus expression QTL). Sign concordance across 354 variants significant in both layers was 0.506 (95% CI 0.455–0.556; binomial $p = 0.87$) — the two layers are statistically independent, with no cross-cell-type confound. A secondary caQTL-versus-enhancer-activity comparison (685 variants, MPRA/SuRE) reproduced this (0.528, 0.491–0.565) and survived removal of estimate-derived datasets. The Alzheimer's-disease risk variant rs6733839 (upstream of *BIN1*) anchors the mechanism: the risk allele opens chromatin (caQTL $Z = 3.3$), raises *BIN1* expression (eQTL $Z = 6.4$), yet represses episomal enhancer activity (MPRA). A Boltz-2 co-fold of the MEF2A dimer on the local enhancer is high-confidence for both alleles (interface pTM 0.978) and shows a tighter protein–DNA interface at the risk allele (+12.5% contacts), consistent with a risk-allele-created MEF2A motif.

Conclusions. Predicting regulatory-variant effect-direction from sequence is limited not by model capacity but by an accessibility–activity decoupling that single-readout models cannot represent. Directional predictions from a model trained on one functional layer should not be interpreted as predictions for another. We release the harmonized benchmark, the paired-layer independence statistic, and all per-variant scores as a reusable community resource.

Keywords: noncoding variants; variant effect prediction; deep learning; chromatin accessibility; expression QTL; microglia; Alzheimer's disease; *BIN1*; benchmark; sequence-to-function models

Background

Genome-wide association studies (GWAS) localize the large majority of disease-associated variants to noncoding sequence, where mechanistic interpretation requires predicting a variant's effect on gene regulation. A decade of sequence-to-function deep learning has transformed this problem: DeepSEA [1] and Basset [2] first showed that convolutional networks trained on ENCODE chromatin data could predict the regulatory impact of single nucleotide changes; the field has since scaled to long-context transformer and U-Net architectures — Borzoi, which predicts RNA-seq coverage and other assays over 524 kb windows [4], and AlphaGenome, which processes up to one megabase at base-pair resolution across eleven molecular modalities [3]. These models advance the state of the art on held-out track prediction and are now routinely used for in-silico variant scoring.

Track-prediction accuracy, however, is not the quantity a variant-interpretation pipeline needs. The clinically and mechanistically actionable question is directional: does a given variant *increase* or *decrease* accessibility, transcription-factor binding, or expression in the relevant cell type? The ground truth for this question comes from functional assays — massively parallel reporter assays, including saturation mutagenesis of regulatory elements at single-base resolution [8], and QTL maps — against which model predictions can be scored. Directional accuracy is nonetheless seldom reported as a headline metric, and independent evaluations on causal-variant benchmarks have found that effect-size correlations for state-of-the-art models remain modest [5]. Whether the leading models can reliably call the *sign* of a variant's effect — particularly for the context-dependent, combinatorial regulatory logic that characterizes disease loci — has not been established on a like-for-like, held-out benchmark.

We addressed this in a single disease-relevant setting: the microglia/brain/immune context, motivated by the *BIN1* Alzheimer's-disease locus and its lead noncoding variant rs6733839. We adopted a benchmark-first discipline throughout: assemble a held-out test set of variants with laboratory-*measured* direction; split by chromosome to prevent linkage-driven leakage; score the leading models against floor baselines and in-distribution controls on the *identical* set; report bootstrap confidence intervals rather than point estimates; and treat any apparent improvement as a potential leakage artifact until excluded. The result reported here is, deliberately, a negative one with a mechanistic explanation. We find that effect-direction is not predictable from local sequence at any model capacity we tested; we show *why* through a paired-layer decoupling analysis in matched donors; and we anchor the mechanism on a single, deeply characterized variant.

Methods

Benchmark assembly and direction convention

We harmonized noncoding variants with measured effect direction from public sources into a single convention: **direction = sign of the effect attributed to the alternate allele in the variant's cell context.** Sources were (i) a 2025 context-dependent Alzheimer's-disease massively parallel reporter assay (MPRA) covering THP-1 macrophage, HMC3 microglia, brain tissue, and HEK293T [13]; (ii) SuRE raQTL data for K562 and HepG2 [7]; (iii) a hiPSC neural-progenitor MPRA (GEO GSE244011); and (iv) a 3′-UTR MPRA in microglia and SH-SY5Y (GEO GSE253841). The assembled matrix comprises 6,377 variant×context measurements across nine contexts. Coordinates were harmonized to GRCh38 (hg19 sources lifted with UCSC liftOver); allele orientation was reconciled to the reference/alternate convention and strand-corrected where required.

On-target chromatin and expression layers

A native human-microglia caQTL map (Kosoy et al. [6]; meta-analysis over 95 donors; AD Knowledge Portal / Synapse syn30308248) contributed measured chromatin-accessibility direction; the matched microglia meta-eQTL map from the same donors (Synapse syn30308484) contributed measured expression direction. For each variant we retained the strongest association (maximum |Z|) and harmonized the reported effect allele to our reference/alternate convention.

Chromosome split

All evaluation uses a fixed split by **chromosome** (test = chr 2, 6, 11, 19; chr2 forced to retain *BIN1*/rs6733839; random seed 0), never by variant, to prevent linkage-disequilibrium leakage between train and test. The split manifest is version- controlled with the data.

Models, baselines, and controls (all scored on the identical held-out set)

- **AlphaGenome** [3] via the public API: 1 Mb context; ATAC and DNase modalities;

DIFF_LOG2_SUM variant scorer; predicted direction = sign of the mean effect over biosample tracks matched to the target context by a curated regular expression.

- **Borzoi** [4] (open weights, replicate 0): 524 kb input windows; predicted direction =

sign of alt–ref over the five central prediction bins of microglia/macrophage accessibility tracks; executed on an NVIDIA L40S GPU.

- **Floor baseline 1 — motif PWM Δ-score:** sign of the largest-magnitude change in JASPAR

[11] motif log-odds score at the variant (49-PWM curated vertebrate panel).

- **Floor baseline 2 — logistic-on-motifs:** L2-regularized logistic regression on

per-motif Δ features, fit on training folds only.

- **In-distribution controls (caQTL set):** logistic regression and gradient-boosted trees

trained directly on the caQTL training folds — an upper bound on the directional signal extractable from local sequence features.

Statistics

Direction accuracy and AUROC are reported with 95% bootstrap confidence intervals (≥2,000 resamples; seed 0). Between-layer sign concordance is tested against 0.5 with an exact binomial test and 5,000-resample bootstrap intervals. A "win" was defined a priori as a bootstrap CI non-overlapping with both incumbents and both floor baselines in at least one named context.

Layer-decoupling analysis

Each variant's microglia caQTL direction was paired with (a) its microglia eQTL direction in the same 95 donors (primary, confound-free) and (b) its enhancer-activity direction from MPRA/SuRE (secondary). Concordance was computed on variants significant in both layers ($|caQTL\ Z| > 1.96$; activity FDR < 0.05 or $p < 0.01$; $|eQTL\ Z| > 1.96$). A drop-one sensitivity analysis removed estimate-derived activity datasets (those for which we re-estimated direction from public count matrices rather than using the original study's model).

Predicting decoupling

We trained an L2 logistic model to predict, per variant, whether the caQTL and activity signs disagree, using effect magnitudes, caQTL significance, cell lineage, and 49 motif-Δ features, evaluated under the same chromosome split (AUROC with bootstrap CI).

Structural co-fold

The MEF2A MADS-box dimer (sequence from PDB 1EGW) was co-folded with the 17-bp *BIN1* enhancer duplex centered on rs6733839, for both alleles, using Boltz-2 [12] with five diffusion samples, three recycling steps, and MSA-server profiles. The protein–DNA interface was quantified as the number of protein–DNA atom pairs within 4 Å.

Results

No model predicts accessibility direction in the native cell type

On the native microglia caQTL test set (361 held-out variants) — the fairest test either frontier model can be given, since chromatin accessibility is precisely what they are trained to predict, in the exact cell type of interest — every method sits at chance (Figure 1; Table 1). AlphaGenome reached 0.537 (95% CI 0.485–0.587), Borzoi 0.551 (0.499–0.601), the motif floor 0.493, and none cleared the 0.565 majority-class rate. Borzoi's run had no zero-inflation (all 361 variants fell inside microglia accessibility peaks), confirming an uncontaminated read. Decisively, models trained *in-distribution* on the caQTL data itself remained at chance (logistic 0.529; gradient-boosted trees 0.499): the directional signal is not present in the local sequence, independent of model capacity. No method met the pre-registered win criterion. This is a well-powered negative result.

Table 1. Direction accuracy on the native human-microglia caQTL test set (n = 361).

Method	Accuracy (95% CI)	AUROC
Motif PWM Δ-score (floor)	0.493 (0.443–0.543)	—
Logistic-on-motifs (in-distribution)	0.529 (0.479–0.579)	0.544
Gradient-boosted trees (in-distribution)	0.499 (0.446–0.549)	0.501
AlphaGenome (1 Mb context)	0.537 (0.485–0.587)	0.547
Borzoi (524 kb context)	0.551 (0.499–0.601)	0.562
Majority-class baseline	0.565	—

Chromatin-accessibility and expression effect-directions are statistically independent

To explain the negative, we tested whether accessibility direction predicts expression direction in the same cells. Using the microglia caQTL and eQTL maps from the *same 95 donors* — eliminating any cross-cell-type confound — sign concordance among 354 variants significant in both layers was 0.506 (95% CI 0.455–0.556; binomial p = 0.87), indistinguishable from 0.5 (Figure 3; Table 2). Every stratum (all-paired, caQTL- significant, eQTL-significant) fell on 0.5. The secondary caQTL-versus-enhancer-activity comparison (685 variants significant in both layers) reproduced this at 0.528 (0.491–0.565; p = 0.15), and the value was stable at ~0.53 when estimate-derived datasets were dropped (paper-statistic-only subset: 0.530; Figure 2). A variant's chromatin- accessibility direction therefore carries essentially no information about its expression/activity direction. We are precise about the nature of the finding: the layers are statistically *independent*, not systematically *anti*-correlated. Independence is sufficient to explain the directional-prediction failure — a model reading one layer cannot recover the sign of another — but it is a quieter claim than a systematic sign-flip, and we report it as such.

Table 2. Same-donor microglia caQTL–eQTL sign concordance, by significance stratum.

Stratum	n	Concordance (95% CI)	Binomial p
All paired (no significance filter)	1,322	0.511 (0.485–0.538)	0.43
Significant in both layers	354	0.506 (0.455–0.556)	0.87
caQTL-significant	617	0.519 (0.480–0.558)	0.38
eQTL-significant	744	0.503 (0.466–0.538)	0.91

Decoupling is not predictable, and the model-failure link is directional but underpowered

A logistic model on effect magnitudes, caQTL significance, lineage, and 49 motif-Δ features failed to predict which variants decouple (chromosome-split AUROC 0.441, 95% CI 0.361–0.524) — an honest negative that precludes offering a decoupling classifier as a tool. Stratifying model accuracy against activity direction by concordance showed the hypothesized shape — both models more accurate on concordant variants (AlphaGenome 0.594, Borzoi 0.625) than on decoupled ones (both 0.538) — but the confidence intervals overlap, so this stratified effect is suggestive rather than significant at present sample size (Figure 2).

rs6733839 anchors the mechanism across three measured layers

The lead *BIN1* Alzheimer's variant rs6733839 — fine-mapped as the credible causal variant at this locus [9] and previously implicated in myeloid regulatory function [10] — is present in all three microglia layers with strong, measured signals (Table 3). The risk allele opens chromatin and raises *BIN1* expression — the two native-genome readouts agree — while repressing episomal enhancer activity in the reporter assay. Both AlphaGenome and Borzoi call one layer correctly and the other backwards, each in a different direction; there is no single model that is "right" about this variant, precisely because the layers it reads disagree with the layer against which it is scored.

Table 3. rs6733839 (Alzheimer's risk allele, T) across measured microglia layers.

Layer	Measured effect of risk-T	Evidence
Chromatin accessibility (caQTL)	opens (+)	Kosoy et al., Z = 3.3
Gene expression (eQTL, *BIN1*)	up (+)	Kosoy et al., Z = 6.4
Episomal enhancer activity (MPRA)	represses (–)	2025 AD-MPRA

A structural co-fold shows a tighter MEF2A interface at the risk allele

Boltz-2 co-folds of the MEF2A dimer on the *BIN1* enhancer were high-confidence for both alleles (interface pTM 0.978; complex pLDDT 0.98), indicating a well-defined complex (Figure 4). The risk-allele complex made more protein–DNA contacts than the non-risk complex (287 versus 255 within 4 Å; +12.5%), consistent with the sequence-level prediction that the risk allele creates a stronger MEF2A motif (+7.56 bits by JASPAR) and with the measured chromatin-opening and expression increase. The transcription factor mediating the *repressive* reporter-assay effect remains unresolved: the originating AD-MPRA study attributes it to SPI1, whereas our sequence and structural evidence favor a MEF2A site-creation event. We report both interpretations and force neither.

Discussion

Across every context we assembled — neural-progenitor MPRA, 3′-UTR MPRA, and a native microglia caQTL map — the direction of a noncoding variant's regulatory effect is not predictable from sequence, by frontier models, by floor baselines, or by models trained in-distribution. The consistency across contexts, together with the in-distribution control, argues that this is a property of the prediction problem rather than of any particular model's capacity.

The paired-layer analysis supplies a mechanism. Chromatin accessibility and gene expression respond to a variant in statistically independent directions (same-donor concordance 0.506). A sequence model that predicts one functional layer therefore cannot, even in principle, recover the sign of another layer on which it was not trained — and the disease-relevant readout, expression, is frequently the one it is not reading. rs6733839 renders the abstraction concrete: a single variant that opens chromatin, raises expression, and yet represses an isolated enhancer fragment, so that the "correct" model depends entirely on which assay one scores against.

The practical implication for variant-interpretation pipelines is cautionary. Accessibility- direction predictions from sequence models should not be read as expression-direction predictions; the two are empirically decoupled, and conflating them will produce confidently wrong directional calls on exactly the context-dependent variants that matter most for disease. This does not diminish the value of these models for track prediction or for prioritizing which variants are regulatory; it bounds their use for calling the *direction* of effect on a specific downstream readout.

Two directions would strengthen the claim. First, a larger same-cell-type paired set would give the model-failure-by-concordance stratification the power to reach significance. Second, orthogonal single-locus validation (allele-specific transcription-factor binding or a reporter titration at rs6733839) would elevate the MEF2A mechanism from a sequence-and-structure prediction to direct evidence. Both are natural extensions; neither is required for the central, well-powered result reported here.

Conclusions

Effect-direction of noncoding regulatory variants is not recoverable from local sequence by current models, and this is explained by a statistical independence between a variant's chromatin-accessibility and gene-expression effects rather than by insufficient model capacity. We provide a harmonized, chromosome-split benchmark with measured direction across nine contexts, a matched-donor decoupling statistic, a full set of per-variant scores for two frontier models and floor baselines, and a structurally anchored case study at the *BIN1* Alzheimer's locus. We release these as a reusable resource for the community and as a cautionary reference point for directional variant interpretation.

Limitations

We state these plainly, as they bound the claims above.

1. The headline is that the layers are statistically *independent* (concordance ≈ 0.5), not

that they systematically flip; this explains the prediction failure but is a weaker claim than a directed anti-correlation, which we do not make.

2. The model-failure-by-concordance link has the predicted direction but overlapping

confidence intervals; a larger same-cell-type paired set is needed for significance.

3. Decoupling is not itself predictable from the sequence features we tested (AUROC 0.441),

so we do not offer a decoupling classifier.

4. There is no wet-lab validation; the MEF2A site-creation, its thermodynamic

interpretation, and the Boltz-2 co-fold are computational, and the +12.5% interface difference is a single-model geometric readout, not a free-energy calculation.

5. The transcription factor responsible for the reporter-assay repression is unresolved

(SPI1 versus MEF2A).

6. Some activity-layer directions are our re-estimates from public count matrices rather

than the original studies' models; the core result is robust to dropping these, and they are flagged in the provenance record.

7. Some benchmark variants may lie within the incumbent models' training data (ENCODE/GTEx);

this can only inflate their scores, and they remain at chance.

Declarations

Availability of data and materials

All data, split manifests, per-variant model scores, figures, and analysis code supporting the conclusions of this article are available in the repository `github.com/anujdevsingh/regulatory-variant-agent` under the `bench/` and `results/` directories, committed per analysis stage with configuration and random seed. Primary public datasets: human-microglia caQTL and meta-eQTL summary statistics (Kosoy et al. [6]; AD Knowledge Portal / Synapse syn30308248 and syn30308484), which are registered-access — freely downloadable after creating an account and agreeing to the portal's data-use terms, with no sponsor approval required; SuRE raQTL [7]; the 2025 context-dependent AD-MPRA [13]; GEO GSE244011 and GSE253841 (open). Structure predictions used Boltz-2 [12]; variant scoring used the AlphaGenome API [3] (non-commercial terms) and Borzoi open weights (replicate 0) [4]. Genome build GRCh38.

Ethics approval and consent to participate

Not applicable. This study is a secondary analysis of previously published, de-identified public datasets and does not involve new human or animal subjects.

Consent for publication

Not applicable.

Competing interests

The author declares no competing interests.

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[To be completed by the author.]

Authors' contributions

A.D.S. conceived the study, assembled the benchmark, performed all analyses, produced the figures, and wrote the manuscript.

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Figure legends

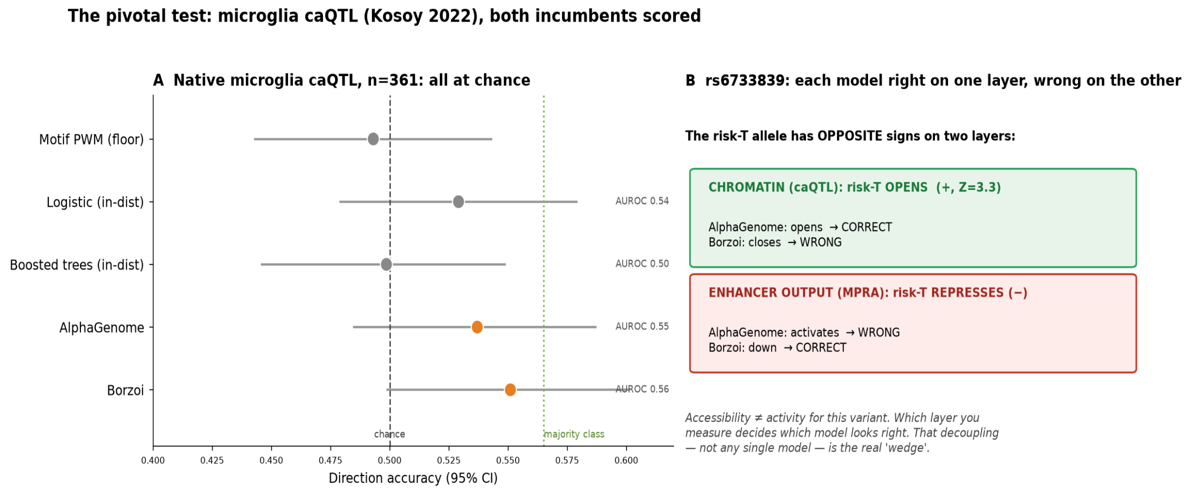

Figure 1. *Direction accuracy on the native human-microglia caQTL test set (n = 361). Both frontier models (AlphaGenome, Borzoi), motif floor baselines, and in-distribution controls fall at or below the majority-class baseline (0.565); error bars are 95% bootstrap confidence intervals.*

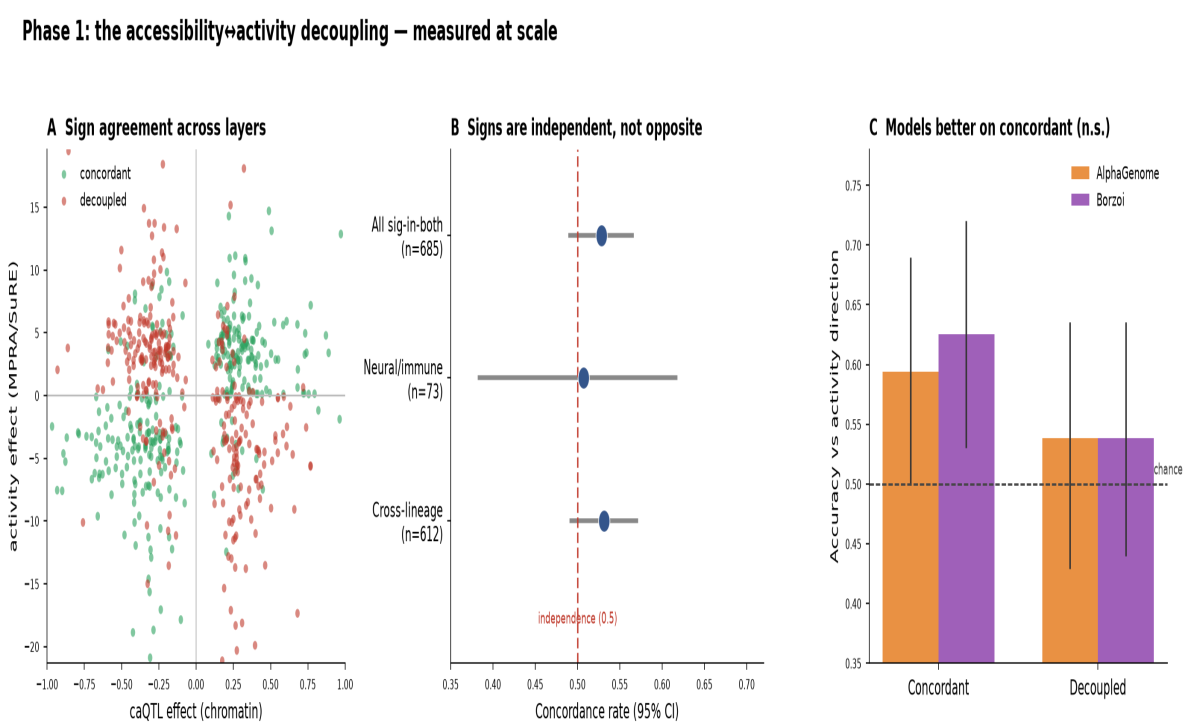

Figure 2. *Layer-decoupling at scale (caQTL versus MPRA/SuRE enhancer activity). Sign agreement by stratum, robustness to dropping estimate-derived datasets, and model accuracy stratified by concordant versus decoupled variants.*

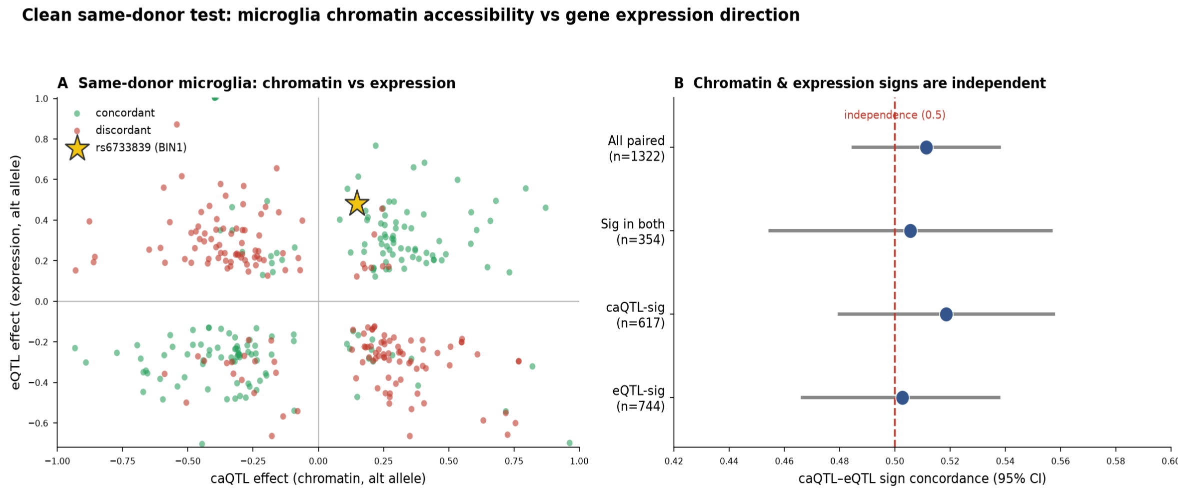

Figure 3. *Clean same-donor test: microglia caQTL versus eQTL. (A) Per-variant effect sizes coloured by concordance, with rs6733839 (BIN1) marked. (B) Sign concordance with 95% confidence intervals across significance strata; all strata are indistinguishable from independence (0.5).*

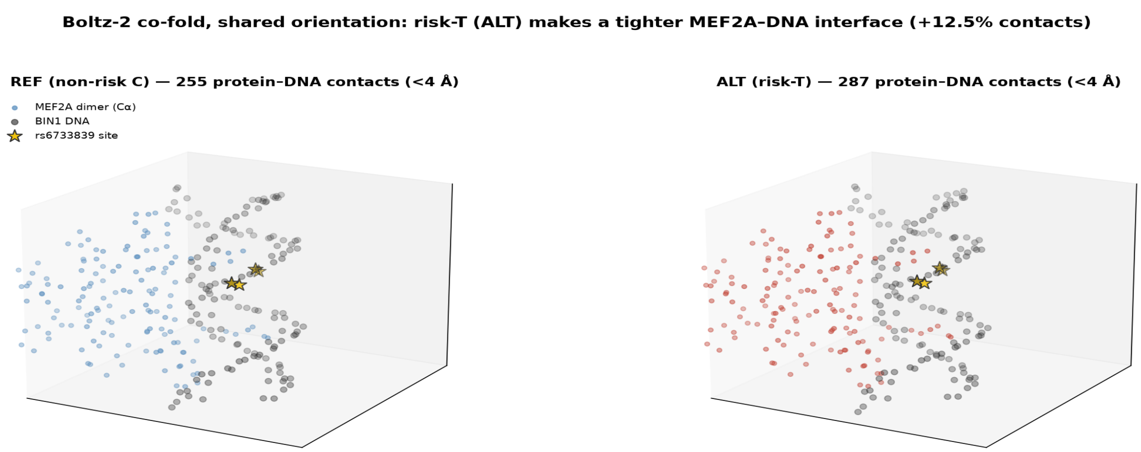

Figure 4. *Boltz-2 co-fold of the MEF2A dimer on the BIN1 enhancer, shown in a shared orientation for the reference (non-risk C) and alternate (risk T) alleles. Both complexes are high-confidence (interface pTM 0.978); the risk allele forms a tighter protein–DNA interface (+12.5% contacts within 4 Å). The rs6733839 site is marked.*